Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 204–214

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Enhancement of Multiclass Dermoscopic Image Classification with a Fusion Model and Fine-Tuned Deep Neural Networks

Sonam Khattar[a*], Dr. Rohit Bajaj[b]

[a*]*Department of Computer Science, Chandigarh University, Gharuan, Mohali, Punjab 140413, India*
[b]*Department of Computer Science, Chandigarh University, Gharuan, Mohali, Punjab 140413, India*
sonam24jun@gmail.com [a] , rohit.rick@gmail.com [b]*

Abstract

Since malignancy of the skin is a frequent and sometimes fatal illness, prompt and precise detection is essential to successful treatment. Dermoscopic imaging is now emerging as a helpful tool for dermatologists in aiding with the identification of various types of skin lesions and offering timely treatments. Due to varied image conditions, data unbalancing, and fine-grained pattern classification task are is challenging in skin cancer detection. Dermoscopic images vary greatly in quality, illumination, and orientation, making it challenging for models to efficiently generalize across varied image conditions. To obtain accurate and reliable skin cancer detection, such challenges require the utilization of sophisticated deep learning techniques and rigorous model development. In multiple classes skin malignancy classification, convolutional neural networks outperform dermatology professionals, according to recent research. In the proposed system, firstly a preprocessing technique has been developed that includes hair and noise removal, data augmentation, and image resizing with model parameters. Then training has been performed using transfer learning on the fusion of DenseNet121 and ResNet50V2 model on the HAM10000 dataset. The proposed technique's performance is assessed using measures such as sensitivity, f-1 score, specificity, and accuracy. For multi-class skin lesion classification, the suggested technique achieves a precision of 97.88%, F-Score of 98%, sensitivity of 99%, accuracy of 97.08%, and Mathew's correlation coefficient (MCC) of 96.4%, it executes superior to cutting-edge techniques.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Dermoscopy, Skin lesion, Hybrid, transfer learning, DenseNet, ResNet, Classification

1. Introduction

The development of skin cancer has emerged as a significant global health issue, with an unprecedented rise in its occurrence in recent years. Based on the most recent statistics available, skin cancer ranks as one of the

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.210

malignancies that is now diagnosed most commonly worldwide. According to estimates from the United States Cancer Society, there are two to three million cases of non-melanoma skin cancer annually, with approximately 104,930 instances of new cases and 12,470 instances of estimated deaths [1]. These figures highlight the critical need for new techniques of early diagnosis and precise classification to reduce the disease's effect on public health. The increasing incidence of skin cancer requires the development of novel techniques for early diagnosis and accurate classification [2]. Dermoscopic imaging has emerged as a useful method for improving the accuracy of skin cancer diagnosis by allowing doctors to visualize skin lesions at a microscopic level. Nevertheless, the manual evaluation of dermoscopic images requires a lot of time and is subject to human error. Consequentaly, there is a rising interest in using sophisticated computational approaches, including deep learning, to construct automated and trustworthy skin cancer classification systems. It can help healthcare professionals discover skin cancer at an early stage by offering a proposed diagnosis, thereby lowering patient mortality and treatment costs [3]. Skin lesion images from the HAM10000 dataset is shown in fig 1.

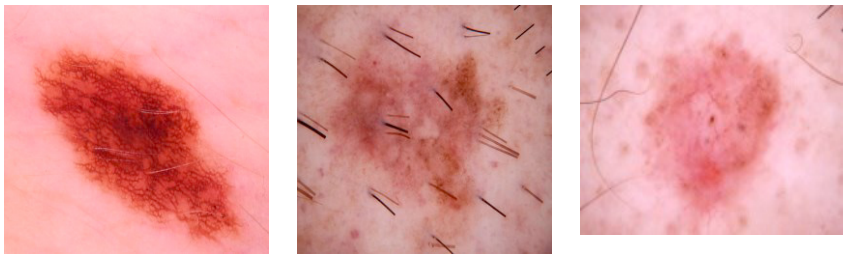


Fig.1. Skin lesion images from the HAM10000 dataset

The primary purpose of this study was to create an advanced skin classification model asing dermoscopic images. Through the use of deep transfer learning, this study aims to improve the efficacy as well as accuracy of dermatological cancer diagnosis, especially by combining the strengths of the DenseNet and ResNet architectures on the HAM10000 dataset. By utilizing these sophisticated approaches, we intend to provide physicians with a helpful tool that may assist in the timely detection of skin cancer categories, resulting in more prompt treatments and improved patient outcomes. The primary contribution to this research are as follows:

1) A preprocessing approach, such as hair and noise removal, augmentation of data, and resizing of image were employed to enhance data quality for robust skin cancer classification.
2) Transfer learning approach that successfully captures fine-grained patterns for precise skin lesion classification by fuzing DenseNet121 and ResNet50V2 models.
3) Outperformed existing standards in precision (97.88%), F-measure (98%), MCC (98%), Recall (97.86%), and accuracy (97.08%) in multi-class skin lesion classification.
4) Outperformed existing state-of-the-art methods, demonstrating deep learning's potential to improve dermatologists' diagnostic capabilities and promote faster treatment solutions.

The remaining sections of the work are arranged as follows: The linked work is described in Section 2, and the planned work is presented in Section 3. Section 4 explores into the result analysis, and Section 5 provides a comprehensive study of the performance of our suggested technique.

2. Related Work

Recent studies have explored several deep Learning (DL) approaches for studying skin problems using dermoscopic images. Despite the imaging system's potential for lesion enhancement, complicated lesion structures present major visual inspection challenges. Automatic skin disease segmentation and classification approaches have been developed to solve this problem. A survey of the literature reveals a various of methodologies, including segmentation, classification, and hybrid methods, some of which involve pre-processing.

A fully computerized method for multi-class skin tumor classification and segmentation using discriminative deep

features was presented by Khan MA et al. in 2021. LCcHIV was used for initial image enhancement, followed by deep saliency segmentation utilising a bespoke CNN. A pre-trained CNN is used to segment color lesion images and extract characteristics. IMFO picks critical traits that are fuzed with MMCA and categorized by KELM. Across the ISBI 2016, PH2, ISBI2017, and ISIC 2018, the suggested technique obtains excellent segmentation accuracy (95.38% to 98.70%), while classification reached 90.67% on the HAM10000.

Alam TM et al. /2022 suggested a framework, as well as hyperparameter adjustments were examined, on a combination of balanced and unbalanced data points, RegNetY-320 beats InceptionV3 and AlexNet in terms of precision, F1-score, and ROC spectrum performance. When compared with conventional approaches, the framework outperformed the of accuracy (91%), F1-score (88.1%), and ROC curve value (0.95). With a curve with a ROC value of 0.90, 85% accuracy, and 69.3% F1-score, the most sophisticated method was exceeded by these results.

The DSCC_Net approach to categorizing the four types of skin cancer diseases was presented by Tahir M et al. in 2023. The model incorporates a modified convolutional block layout with many layers, with the goal of detecting the initial stages skin cancer. The SMOTE Tomek method creates balanced samples for each class to overcome dataset imbalance. Grad-CAM visualization highlights class activation in the CNN layer, demonstrating the model's effectiveness. The DSCC_Net model has notable performance indicators: 94.28% precision, 93.93% F1-score, 93.76% recall, 94.17% accuracy, and an AUC of 99.42%.

Venugopal V et al./2023 employed modified EfficientNet-B4 and EfficientNetV2-M DNN models to categorize lesions on the skin from 58,032 dermoscopic pictures. Generalization is improved via transfer learning and data augmentation. For multiclass and binary classification problems, EfficientNetV2-M outperforms other models with an AUC score of 0.9807 and precision of 95.5%. [4].

Wu QE et al./2023 presented a unique skin pathological mirror classification approach for melanoma skin cancer diagnosis. It solves difficulties such as feature information loss during downsampling and channel information underutilization. To rebuild disease characteristics, the technique employs wavelet downsampling and a multichannel attention mechanism. The experimental findings on the HAM10000 dataset show that the technique outperforms prior techniques in terms of precision (95.84%), noise immunity, and classification speed. Moreover, Table 1 provides an overview of the associated work summaries.

Table 1. Summarizes the related work of recent studies

Ref.	Author/Year	Models	Dataset	Methodology	Performance Matrices
[4]	Khan MA et al./2021	(LCcHIV) local color-controlled histogram intensity values with CNN	ISBI 2016, ISBI 2017 ISIC 2018 PH2	Automated approach for classifying multi-class skin lesions	Accuracy- 90.67% Sensitivity- 90.20%
[5]	Alam TM et al. /2022	RegNetY-320	HAM10000	Skin cancer classifier based on deep learning	Accuracy -91% F1-Score -88.1%
[6]	Tahir M et al./2023	DSCC-Net	HAM1000	Deep Learning Models for Multi-Classification	Accuracy -94.17% Sensitivity- 0.83 Specificity- 0.99 Precision- 94.28% Recall- 93.76% AUC score -99.43
[7]	Venugopal V et al./2023 [4]	modified EfficientNet-B4 and EfficientNetV2-M DNN	ISIC 2020 ISIC 2019 HAM10000	multiclass and binary classification of skin images	Accuracy -95.5% Sensitivity- 0.83 Specificity- 0.99 Precision- 0.83 Recall- 0.94 AUC score -0.9807
[8]	Wu QE et al./2023 [7]	wavelet downsampling with ResNet-50	HAM 10000	classification of skin cancer has been performed by wavelet analysis, which is used as down-sampling in CNNs	Accuracy- 95.28% F1 score - 95.39%

3. Proposed Work

The goal of the suggested research approach is to develop DensRes, an integrated deep transfer learning technique for the classification of malignancies of the skin based on dermoscopic images. The HAM10000 dataset was gathered and preprocessed to ensure visual clarity. DensRes combines the capabilities of the DenseNet and ResNet architectures, utilizing transfer learning on pre-trained skin cancer classification models. To maintain useful features during downsampling, a specialized dermoscopic CNN is included, along with a multichannel attention mechanism. Backpropagation is used to optimize the model after it has been trained on a HAM10000 dataset. Performance is assessed using sensitivity, precision, F1-score, accuracy, specificity, and MCC. The suggested method intends to give dermatologists with an accurate, efficient, and practical tool for skin cancer detection, thus benefiting them and enhancing patient care.

3.1 Data Collection and Pre-processing:

The HAM10000 dataset is a widely known and commonly employed collection of 10,015 dermoscopic images intended for research on skin cancer classification. It contains several skin lesion images, each labeled into one of seven classifications: melanocytic lesions, melanoma, actinic keratoses, benign keratosis, basal cell carcinoma, dermatofibroma and vascular lesions representing various skin conditions [9]. To ensure a representative sample of skin lesions observed in real-world clinical settings, the dataset was assembled from multiple sources, including medical centers and dermatological clinics. The dataset was extensively pre-processed before to model training to improve data quality and solve unique problems. To reduce hair and background noise interference, a morphological filter was employed to remove hair artifacts and a mean filter was used to remove background noise, resulting in cleaner and more informative images [10]. The table 2 presents the classwise distribution and description of the HAM10000 dataset for the proposed hybrid model.

Table 2. Class allocation and the HAM10000 dataset overview

Class Name	Description	Total number of images
akiec	Actinic keratoses and intraepithelial carcinoma	327
bcc	Basal cell carcinoma	514
bkl	Benign Keratosis-like Lesions	1099
df	Dermatofibroma	115
mel	Melanoma	1113
nv	Melanocytic Nevi	6705
vasc	Vascular Lesions	142

3.2 Data Augmentation and Class Balancing

One method to improve the dataset and avoid overfitting is data augmentation. Random rotations, horizontal and vertical flips, zooming, and translation were among them. Augmentation added enhanced versions of each image to the dataset, increasing the effective dataset size and enhancing model generalization. Furthermore, class imbalances had to be addressed to ensure that all classes were fairly represented throughout training. To correct imbalances, class balancing strategies such as oversampling minority classes and undersampling majority classes were used, preventing the model from favouring dominant classes and supporting impartial learning. The suggested procedure for classifying skin lesions is flowcharted in Figure 2.

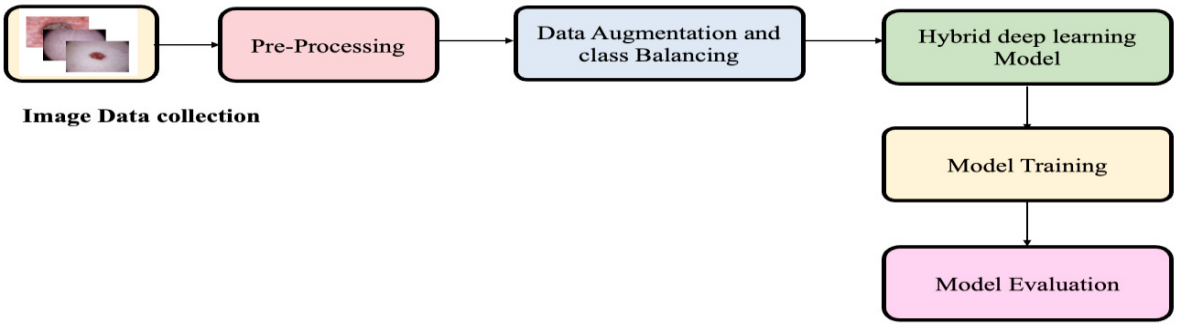


Fig. 2. Flowchart of the proposed Methodology

3.3 Proposed Model Architecture:

The proposed architecture, DensRes, is a hybrid approach for deep transfer learning that blends the capabilities of the DenseNet121 and ResNet50V2 structures related to skin cancer classification using dermoscopic images. TensorFlow-Keras was used to build the model. The input layer has the form (28, 28, 3), which corresponds to the dimensions of the input images. To match the input shape required by DenseNet121 and ResNet50V2, the input images are shrunk to (224, 224, 3) using a custom Lambda layer. DenseNet121 and ResNet50V2 models that have been pre-trained on the ImageNet dataset are loaded without their top classification layers [11]. These models function as feature extractors, extracting rich visual characteristics essential for skin cancer classification. After the feature extraction layers, custom layers are created to minimize the spatial dimensions. A GlobalAveragePooling2D layer is employed for each model, with a ReLU activation function and a dense layer comprising 495 units coming next. To prevent overfitting, a dropout rate of 0.5 was used. A concatenate layer is used to concatenate the outputs of the custom layers from DenseNet121 and ResNet50V2, thereby merging the features retrieved from both architectures [12]. Depending on the complexity of the classification assignment or model needs, additional layers might be added as needed. The final classification layer is a Dense layer that has 7 units (one for each class) using a softmax activation function to generate class probabilities. The two models are then integrated using the concatenated output to create the DensRes model, which takes use of both DenseNet121 and ResNet50V2's capabilities. The layers of both the DenseNet121 and ResNet50V2 models are frozen to prevent the pre-trained layers from being modified during training and to minimise overfitting. The DensRes model is a potent tool for skin cancer classification that usesdermoscopic images that takes advantage of the feature extraction capabilities of both the DenseNet121 and ResNet50V2 architectures, resulting in better classification performance and adaptability. The proposed model architecture is shown in figure 3.

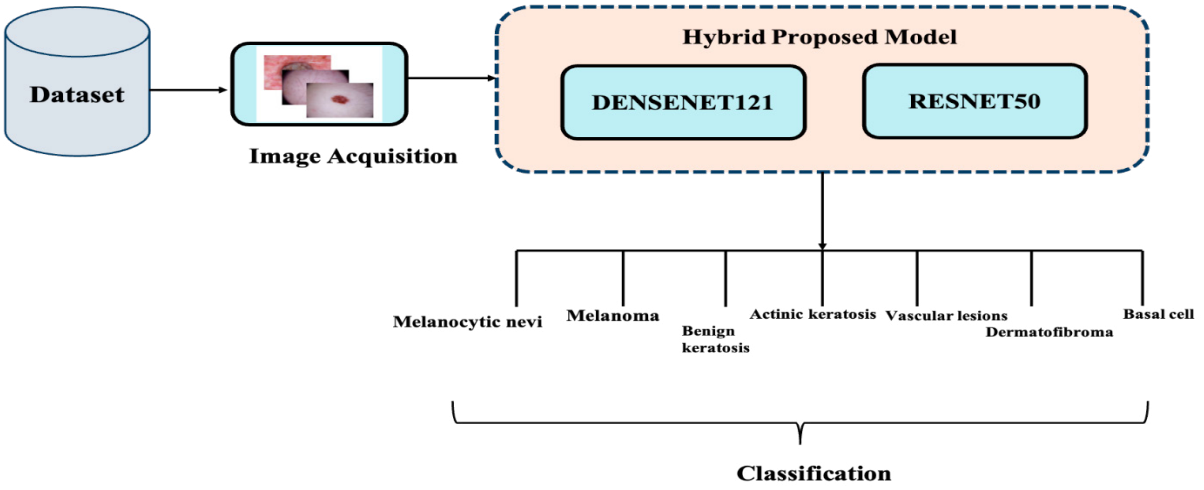


Fig. 3. Proposed model architecture

3.4 Model Training:

To optimize its performance in skin cancer classification, the DensRes architecture is fine-tuned via hyperparameter tuning throughout the model training phase. A crucial hyperparameter that influences the model's convergence and training speed is the learning rate (lr), which is set at $2.4x10^{-4}$. Due to its suitability for multiclass classification, the minimal categorization cross entropy loss function is employed in conjunction with the Adam optimizer during training to optimize the model parameters. The major evaluation parameter is accuracy, It evaluates how well the model can identify lesions under the skin. To optimize training efficiency, a batch size of 34 is used to define the amount of data processed before updating the model parameters. Early stop is also implemented using the ModelCheckpoint and callback, which stores the best model depending on validation correctness. This helps to avoid overfitting and maintains the model with the optimum performance.

Table 3. Hyperparameter configuration

Hyperparameter	Value
Image Size	(224,2224,3)
Learning rate	2.4×10^{-4}
Validation Split	0.2
Number of Epochs	20
Loss function	sparse categorical cross-entropy
Metrices	Accuracy
Batch size	34
Optimizer	Adam

Twenty epochs, or the total number of training iterations over the entire dataset, have been used to train the model. The validation split is set to 0.2, which indicates that 20% of the training data is used for validation during each epoch. Furthermore, Table 3 demonstrates the configuration of the hyperparameter utilized to train the model. This makes it possible to track and assess the model's performance on unobserved data. The model is consistently upgraded during the training process, and its performance is evaluated using the validation accuracy measure. After training, the model will be capable of accurately classifying skin lesions into several skin cancer categories, making it a helpful tool for dermatologists and medical professionals in efficiently identifying and treating skin cancer.

3.5 Model Evaluation:

It is necessary to analyze the suggested DensRes model for identifying skin lesions in order to determine its utility and generalizability. A range of evaluation metrics were employed to appraise the correctness and reliability of the model. To assess the model's performance, many assessment metrics are computed. Sensitivity (recall), precision, F1-score, accuracy, specificity, and the Matthews correlation coefficient (MCC) are the primary performance indicators employed. The model's sensitivity assesses its ability to properly detect positive cases, such as skin cancer, from all real positive cases. Precision is defined as the accuracy of the model's positive predictions expressed as a percentage of all the model's positive predictions. The mean of accuracy and recall, or F1-score, offers a precise evaluation of the model's overall performance. Accuracy measures the model's total proportion of true predictions. Specificity evaluates the model's capacity to discern between extremely negative cases and non-cancerous lesions. An overall connection between model predictions and true labels is provided by the Matthews correlation coefficient, which examines each of the confusion matrix's four regions. These performance measures provides vital insights into the model's strengths and flaws, permitting a comprehensive assessment of its generalization and classification accuracy. The possibilities of the model for increasing skin cancer classification and assisting dermatologists in making informed diagnoses was determined by this evaluation [13].

- Sensitivity (Recall): The precision with which the model can identify positive instances (skin cancer) from all real positive cases.

- Precision: The model's accuracy of positive predictions expressed as a percentage of all the model's positive predictions.
- F1-score: a fair evaluation of the model's effectiveness using the harmonic median of recall and accuracy.
- Accuracy: The model's total percentage of true predictions.
- Specificity: The ability of the model to appropriately distinguish negative instances (non-cancerous lesions) from all genuine negative cases.
- The Matthews correlation coefficient (MCC) assesses the relationship between model predictions and true labels by considering into account all four quadrants of the confusion matrix. The evaluation criteria for model performance is shown in table 4.

Table 4. Evaluation criteria for model performance

Metrices	Description	Mathematic presentation
Sensitivity	The percentage of real positive situations with true positive predictions that measures the model's ability to accurately detect skin cancer patients.	$\frac{TP}{TP+TN}$
Precision	The fraction of genuine positive predictions made by the framework out of all the positive forecasts, indicating the accuracy of the positive categorization.	$\frac{TP}{TP+FP}$
F1-Score	The mean of accuracy and recall was used to assess the ultimate efficacy of the model in identifying skin cancer cases.	$2\times\frac{precision\times recall}{precision+recall}$
Accuracy	A proportion of the model's accurate predictions to its overall predictions, which shows how accurate the model's classifications are overall.	$\frac{TP+TN}{TP+TN+FP+FN}$
Specificity	The proportion of accurate negative predictions among all real negative instances, which assesses the accuracy of the model in identifying benign lesions.	$\frac{TN}{TN+FP}$
Matthews correlation coefficient (MCC)	A correlation coefficient that considers into account each of the confusion matrix's four quadrants and offers a broad assessment of the model's classification efficiency.	$\frac{TN\ \times TP-FN\times FP}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$

4. Results Analysis

This section includes a detailed examination of the findings from the experimental testing of the proposed DensRes model for the classification of lesions in the skin. The consequences of these findings, their importance in the context of skin cancer diagnosis, and potential limitations are discussed. A wide range of assessment criteria, such as accuracy, recall, F1-score, specificity, and the Matthews correlation coefficient (MCC), were used to properly analyze the DensRes model's performance. Table 5 summarizes the results obtained for several skin lesion classifications. This table summarizes the comprehensive performance metrics for the DensRes model's classifications. The model's accuracy in classifying different kinds of skin tumors is demonstrated by the data, which also highlight its shortcomings.

Table 5. Performance metrics for the proposed hybrid model

Class	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)	MCC
Melanocytic nevi (NV)	97.58	99.84	99.68	99.76	99.85	0.996
Benign keratosis (BKL)	95.78	96.38	89.96	93.08	96.91	0.879
Melanoma (Mel)	99.04	99.27	99.48	99.38	99.84	0.986
Actinic Keratosis (AKIEC)	98.36	95.31	95.79	95.55	98.94	0.902
Basal cell carcinoma (BCC)	99.42	99.71	100.0	99.85	100.0	0.998
Dermatofibroma (DF)	99.85	99.24	100.0	99.62	100.0	0.993
Vascular lesions (VASC)	99.85	99.70	100.0	99.85	100.0	0.998

Following 20 epochs of DensRes model training, we found encouraging performance measures that verifed the capacity of the model to categorize skin lesions. The training method resulted in a training accuracy of 97.56%, that the training data's patterns were effectively recognized by the model through training. The model had a validation accuracy of 97.08% during validation, indicating its capacity for effectively extrapolation to previously encountered dermoscopic images. During training the difference between the expected and actual labels is measured by the loss function, was reduced to 0.0833. Similarly, the validation loss attained a value of 0.0977, indicating that the model's predictions on both training and validation datasets were consistently near to the true label. The envisioned work's training curves and validation loss and precision graphs is shown in figure 4.

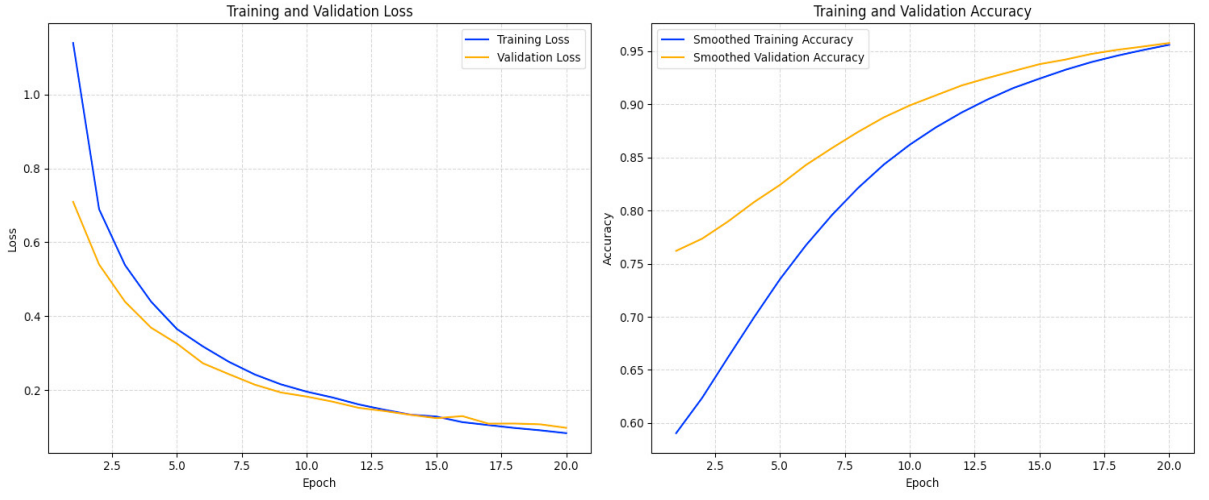


Fig. 4. The envisioned work's training curves and validation loss and precision graphs

In Figure 5, the confusion matrix depicts the classification model's performance across multiple skin lesion classifications. The rows are real classes, and the columns are expected classes. The significant values along the diagonal presents that the model excels at discriminating "Melanocytic nevi (NV)" and "Vascular lesions (VASC)" with high accuracy. The matrix does, however, show certain issues, such as the model's sometimes misunderstanding between the "Melanoma (Mel)" and "Benign Keratosis (BKL)" classes, as well as between the "Melanoma (Mel)" and "Actinic Keratosis (AKIEC)" classes. The high scores in the off-diagnoal cells of these rows indicate some degree of misclassification between these groups. These findings can help influence future model revisions, particularly when it comes to eliminating misunderstandings between closely related categories.

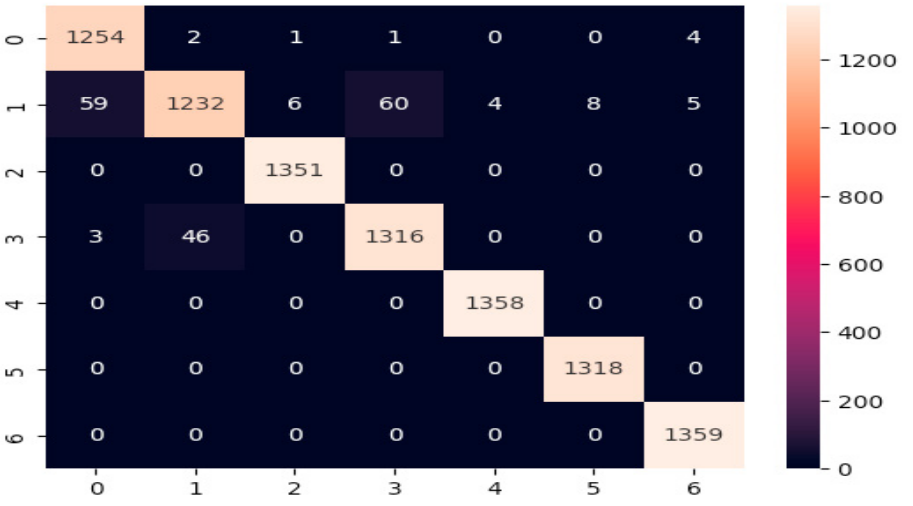


Fig. 5. Confusion matrix of the Classification of proposed approach

The outcomes show how very promising the DensRes model is for classifying skin lesions. The model demonstrates good training and validation accuracies, indicating its ability to capture intricate patterns and characteristics in dermoscopic pictures. This implies that the model effectively used the strengths of the DenseNet121 and ResNet50V2 architectures to improve its classification skills. It is implied that the framework was not overfitted by the validation and training error values' close convergence, signaling that the model's learnt representations expand effectively to newly collected data. Further testing on a separate dataset, on the other hand, is required to gain a more comprehensive evaluation of the model's success in practice.

5. Comprehensive Analysis of the proposed Model:

The comparative analysis provided in Table 6 provides a thorough overview of the performance of multiple skin lesion classification algorithms across various datasets. The efficacy of each model was assessed using important performance parameters such as Matthews Correlation Coefficient (MCC), F1-score, accuracy, precision, and sensitivity. On the HAM1000 dataset, Zhang J et al.'s ARL-CNN attained an accuracy of 88.6%, indicating competitive performance[14] . On the same dataset, Almaraz-Damian JA et al.'s Hand-Crafted CNN achieved an accuracy of 92.4%, demonstrating a balanced approach in collecting both positive and negative cases [15]. Additionally, Datta SK et al.'s algorithm incorporating the Soft-Attention Mechanism attained a HAM10000 dataset accuracy of 93.4%, demonstrating the machine's efficiency in recognizing multiple skin lesion classifications [16]. On the HAM10000, Ali MS et al.'s Deep Convolutional Neural Network attained an accuracy of 91.4%, highlighting its powerful predictive capacity [17]. 87.9% accuracy was achieved by Ali K et al.'s EfficientNet B4 model, high performance with room for improvement in certain criteria [18]. Chaturvedi SS et al. suggested ResNetXt-101demonstrated a performance accuracy of 93.20%, with the possibility to fine-tune some features for even more thorough classification [19] . Sukanya ST et al.'s Deep Belief Net, nevertheless, attained a promising 94.00% accuracy. [15] . On the information set HAM10000, however, our proposed Hybrid DensRes model stood out with an accuracy of 97.08%. This model demonstrated not only high accuracy but also well-balanced precision, recall, F1-score, and sensitivity values over 97%, highlighting its potential for precise and reliable skin lesion classification. Fig 6 elaborated graphically comparison analysis of proposed work.

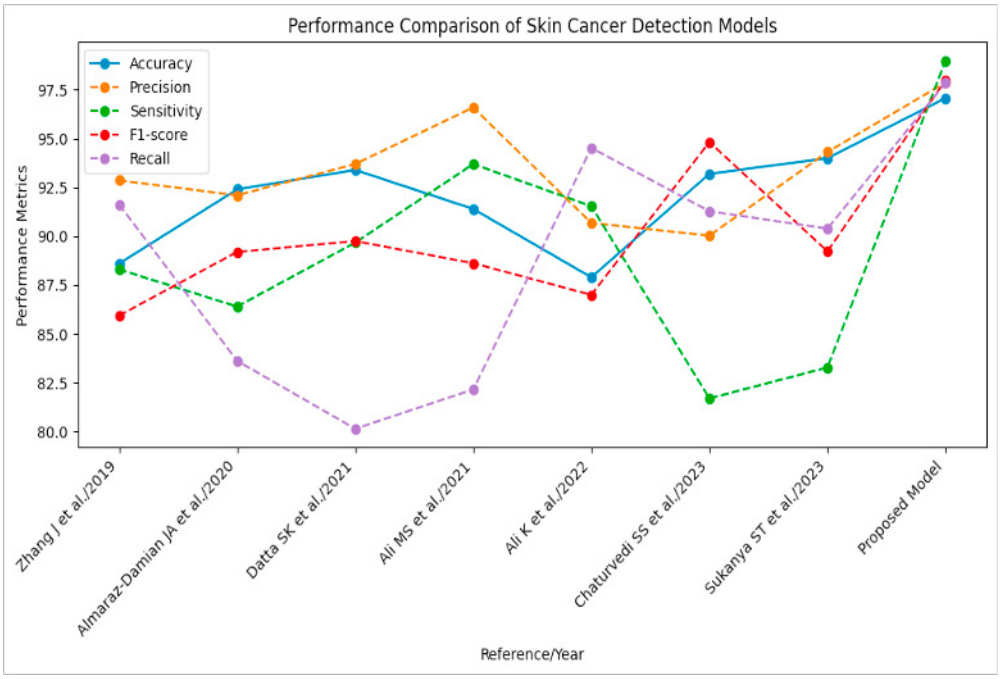


Fig. 6 .Comparison analysis of suggested technique

Furthermore, the comparison analysis in Table 6 of the proposed model highlights the distinct strengths and capacities of different methods for classifying blisters. These results improve skin lesion identification's precision and effectiveness, offering useful insights to the medical community and encouraging further research in this field.

Table 6: Comprehensive Analysis of proposed Model

Refrence/Year	**Model Used**	**Dataset**	**Performance Metrices**
Zhang J et al./2019 [14]	ARL-CNN	HAM1000	Accuracy = 88.6%
Almaraz-Damian JA et al. /2020 [15]	Hand-Crafted with CNN	HAM1000	Accuracy =92.4%. Precision = 92.1% Sensitivity= 86.4% F1-score =89.2%
Datta SK et al./2021[16]	Soft-Attention mechanism	HAM10000	Accuracy =93.4%. Precision=93.7%
Ali MS et al./2021 [17]	Deep Convolutional Neural Network	HAM10000	Accuracy = 91.4% Precision = 96.6% Sensitivity= 93.7%
Ali K et al./2022 [18]	EfficientNet B4	HAM10000	Accuracy = 87.9% F1- Score = 87%
Chaturvedi SS et al. /2023 [19]	ResNetXt-101	HAM10000	Accuracy = 93.20%
Sukanya ST et al./2023 [20]	Deep Belief Net	HAM10000	Accuracy = 94.00%
Proposed Model	**Hybrid DensRes models**	**HAM10000**	Accuracy= 97.08 % Sensitivity: 99% F1-score: 98% Recall = 97.86% Precision:97.88%

6. CONCLUSION

The incidence of carcinoma of the skin is rising, The urgency of a precise and timely diagnosis is growing. This study performed a thorough investigation into improving skin lesion classification using dermoscopic images and sophisticated deep transfer learning models. Developing a reliable skin image categorization model was our main objective by combining the strengths of the DenseNet and ResNet architectures. On the challenging HAM10000 dataset, the Hybrid DensRes models achieved an exceptional accuracy rate of 97.08%. Notably, the model's sensitivity, F1-score, recall, and accuracy scores are consistently above 97%. These findings highlight the model's potential to considerably raise the diagnostic precision of cutaneous lesions. Comparing our methodology with cutting-edge methods validates its efficacy. Its ability to discriminate between distinct skin lesion classifications is extremely valuable to medical practitioners, allowing for early identification and intervention. These discoveries represent a significant step forward in terms of improving patient outcomes and optimizing medical care. Finally, this study contributes to extending the collection of knowledge in the fields of dermatology and deep learning applications. The effective combination of DenseNet and ResNet architectures, together with diligent data preparation and

augmentation, demonstrates the power of sophisticated approaches in medical image analysis. As technology progresses, the prospects for more precise and efficient skin lesion classification algorithms are promising. Such advancement not only advances the area of early cancer diagnosis, but also raises the level of patient care.

References:

[1] Davalos V, Esteller M. Cancer epigenetics in clinical practice. CA: A Cancer Journal for Clinicians. 2023 Jul;73(4):376-424.

[2] Harangi B. Skin lesion classification with ensembles of deep convolutional neural networks. Journal of biomedical informatics. 2018 Oct 1;86:25-32.

[3] Adegun A, Viriri S. Deep learning techniques for skin lesion analysis and melanoma cancer detection: a survey of state-of-the-art. Artificial Intelligence Review. 2021 Feb;54:811-41.

[4] Khan MA, Sharif M, Akram T, Damaševičius R, Maskeliūnas R. Skin lesion segmentation and multiclass classification using deep learning features and improved moth flame optimization. Diagnostics. 2021 Apr 29;11(5):811.

[5] Alam TM, Shaukat K, Khan WA, Hameed IA, Almuqren LA, Raza MA, Aslam M, Luo S. An efficient deep learning-based skin cancer classifier for an imbalanced dataset. Diagnostics. 2022 Aug 31;12(9):2115

[6] Tahir M, Naeem A, Malik H, Tanveer J, Naqvi RA, Lee SW. DSCC_Net: Multi-Classification Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images. Cancers. 2023 Apr 6;15(7):2179.

[7] Venugopal V, Raj NI, Nath MK, Stephen N. A deep neural network using modified EfficientNet for skin cancer detection in dermoscopic images. Decision Analytics Journal. 2023 Jun 24:100278.

[8] Wu QE, Yu Y, Zhang X. A Skin Cancer Classification Method Based on Discrete Wavelet Down-Sampling Feature Reconstruction. Electronics. 2023 May 4;12(9):2103.

[9] Tschandl P, Rosendahl C, Kittler H. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Scientific data. 2018 Aug 14;5(1):1-9.

[10] Khattar S, Bajaj R. Investigation of Denoising Techniques for Removal of Hair and Noise from Dermoscopic Images. In2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS) 2023 Jun 14 (pp. 1190-1197). IEEE.

[11] Huang G, Liu Z, Van Der Maaten L, Weinberger KQ. Densely connected convolutional networks. InProceedings of the IEEE conference on computer vision and pattern recognition 2017 (pp. 4700-4708).

[12] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. InProceedings of the IEEE conference on computer vision and pattern recognition 2016 (pp. 770-778).

[13] Erickson BJ, Kitamura F. Magician's corner: 9. Performance metrics for machine learning models. Radiology: Artificial Intelligence. 2021 May 12;3(3):e200126.

[14] Zhang J, Xie Y, Xia Y, Shen C. Attention residual learning for skin lesion classification. IEEE transactions on medical imaging. 2019 Jan 20;38(9):2092-103.

[15] Almaraz-Damian JA, Ponomaryov V, Sadovnychiy S, Castillejos-Fernandez H. Melanoma and nevus skin lesion classification using handcraft and deep learning feature fusion via mutual information measures. Entropy. 2020 Apr 23;22(4):484.

[16] Datta SK, Shaikh MA, Srihari SN, Gao M. Soft attention improves skin cancer classification performance. InInterpretability of Machine Intelligence in Medical Image Computing, and Topological Data Analysis and Its Applications for Medical Data: 4th International Workshop, iMIMIC 2021, and 1st International Workshop, TDA4MedicalData 2021, Held in Conjunction with MICCAI 2021, Strasbourg, France, September 27, 2021, Proceedings 4 2021 (pp. 13-23). Springer International Publishing

[17] Ali MS, Miah MS, Haque J, Rahman MM, Islam MK. An enhanced technique of skin cancer classification using deep convolutional neural network with transfer learning models. Machine Learning with Applications. 2021 Sep 15;5:1000

[18] Ali K, Shaikh ZA, Khan AA, Laghari AA. Multiclass skin cancer classification using EfficientNets–a first step towards preventing skin cancer. Neuroscience Informatics. 2022 Dec 1;2(4):100034.

[19] Sukanya ST, Jerine S. Skin lesion analysis towards melanoma detection using optimized deep learning network. Multimedia Tools and Applications. 2023 Feb 21:1-23.

[20] Chaturvedi SS, Tembhurne JV, Diwan T. A multi-class skin Cancer classification using deep convolutional neural networks. Multimedia Tools and Applications. 2020 Oct;79(39-40):28477-98.